



DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

University Examination 2025/2026

THIRD YEAR SECOND SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE MECHATRONIC ENGINEERING

EMT 3206: DIGITAL ELECTRONICS I

DATE: 13TH FEBRUARY 2026

11.00AM-1.00PM

TIME: 2 HOURS

Instructions

1. This paper contains FIVE questions
2. Answer Question **ONE** (Compulsory Question), **One** question in **Section A** (Question TWO or Question THREE) and **One** question in **Section B** (Question FOUR or Question FIVE)
3. Remember to indicate all steps and the relevant units in any calculations

Question ONE (Compulsory) [30 marks]

- a) You are required to design a counter for a mechatronic control system that goes through the states "000", "010", "011" and "101". You are to implement the **counter circuit** that achieves the desired sequence using **JK flip-flops**. Develop the **state change diagram**, **state transition table**, **simplified Boolean** [10 marks]
- b) Demonstrate by means of truth tables the validity of **DeMorgan's theorem** for **three variables**:
$$XYZ = \overline{X} + \overline{Y} + \overline{Z}$$
 [2 Marks]
- c) Demonstrate by means of truth tables the validity of the following theorems of Boolean algebra using 3 inputs i.e. (A, B and C).
 - i. The **associative law** [4 Marks]
 - ii. The **distributive law** [4 Marks]
- d) Digital systems were designed as an improvement to traditional analogy devices. Give a formal definition of **digital systems** and state the advantages of digital systems over their **analogue counterparts**. [5 Marks]

$$(A+B)+C$$
$$A+(B+C)$$

$$\begin{array}{cc} Q_n & Q_{n+1} \\ 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{array}$$
$$\begin{array}{cc} J & K \\ 0 & x \\ 1 & x \\ x & 1 \\ x & 0 \end{array}$$

SECTION B

Question FOUR [20 marks]

- a) Using a suitable circuit diagram of a positive-edge triggered, four-bit, **D flip-flop Ring Counter**, explain its operation using timing waveforms and a summary table. Assume the initial state of the circuit as $Q_3, Q_2, Q_1, Q_0 = 1000$ [10 Marks]
- b) You are required to design a 3 bit binary counter using **SR flip-flop** that has the following repeated binary sequence: 0, 1, 2, 3, 4, 5, 6, 7. Generate the state diagram, state transition table, K-map simplification and final circuit implementation. [10 Marks]

Question FIVE [20 marks]

- a) Highlight the 4 key differences between **Combinational** circuits and **Sequential** circuits in digital electronics [4 Marks]
- b) Convert the following numbers to the indicated bases. [6 Marks]
- 7562.45_{10} to **Octal**.
 - 175.175_{10} to **Binary**
 - 1010011.101_2 to **Decimal**
- c) As the digital design engineer for an automotive company, you are required to design a logic circuit to control the vehicle's warning light safety feature. The system consists of 4 input switches and 1 output LED. The switches are labeled D, I, L and S which are used to indicate the status of the driver's Door, the Ignition, the head-Lights and the driver's Seatbelt respectively. The output LED is to light under the following undesired conditions:
- The headlights are ON while the ignition is OFF $\overline{D} \overline{I} L I$
 - The door is OPEN (ON) while the ignition is ON $D I$
 - The seatbelt is UNFASTENED (OFF) while the ignition is ON $\overline{S} I$

Generate the required:

- Truth table** relating the switch inputs and the LED outputs. [3 Marks]
- Using K-Map, **simplify the output equation**. [3 Marks]
- Implement the logic circuit using only **NAND** and only **NOR** gates. [4 Marks]